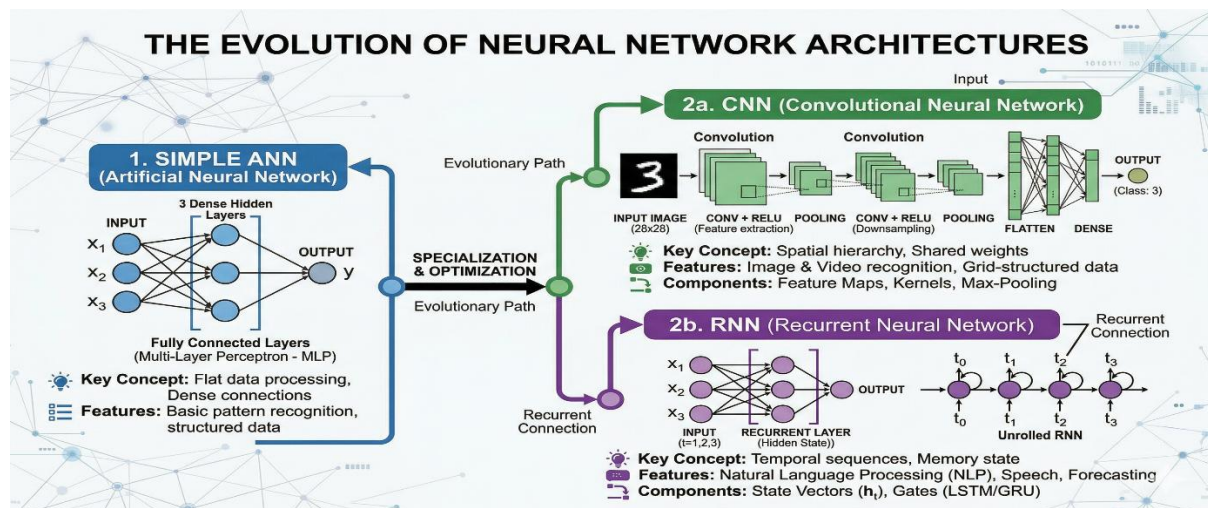


Architecture Comparison Debate

Deep Learning Evolution

Artificial Neural Networks (ANN) are machine learning models inspired by the human brain, consisting of interconnected nodes called neurons. While a standard ANN has three main layers—Input, Hidden, and Output—modern problems require more specialized structures. These advanced architectures, namely CNN, RNN, and GAN, extend the basic feedforward logic to handle specific data types like images, sequences, and generative tasks.



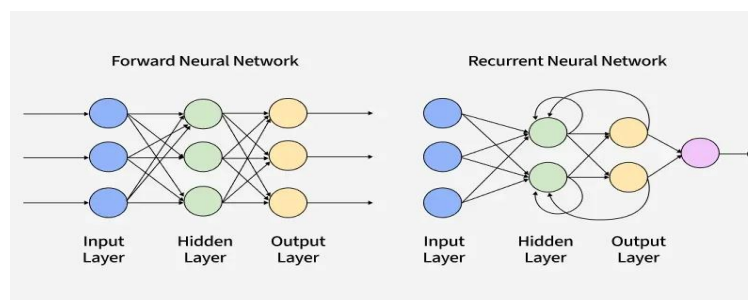
Core Architectural Definitions

- **CNN (Convolutional Neural Network):** Designed specifically for processing pixel data and spatial hierarchies.
- **RNN (Recurrent Neural Network):** Designed for sequential data where the output depends on previous computations.
- **GAN (Generative Adversarial Network):** A framework where two neural networks compete against each other to create new data.

Specific Layer Comparisons

In a standard neural network, the hidden layers process data linearly. However, these specialized architectures change how data moves through the "Hidden" section:

- CNN Structure: Uses Convolutional layers to identify patterns (edges, shapes) and Pooling layers to reduce spatial size.
- RNN Structure: Contains feedback loops that allow information to persist, essentially giving the network a "memory".
- GAN Structure: Consists of a Generator (creating data) and a Discriminator (evaluating data) working simultaneously.



Mathematical Foundation

Just as an ANN uses $Z = (W \times X) + B$ to calculate values, these models have specific formulas.

. CNN Convolution: $O = (I \times K) + B$

(Where I is input and K is kernel and B is Bias)

. RNN State: $h_t = f(W_h \times h_{t-1} + W_x \times x_t + b)$

(Where h_t is the current state and h_{t-1} is the previous memory)

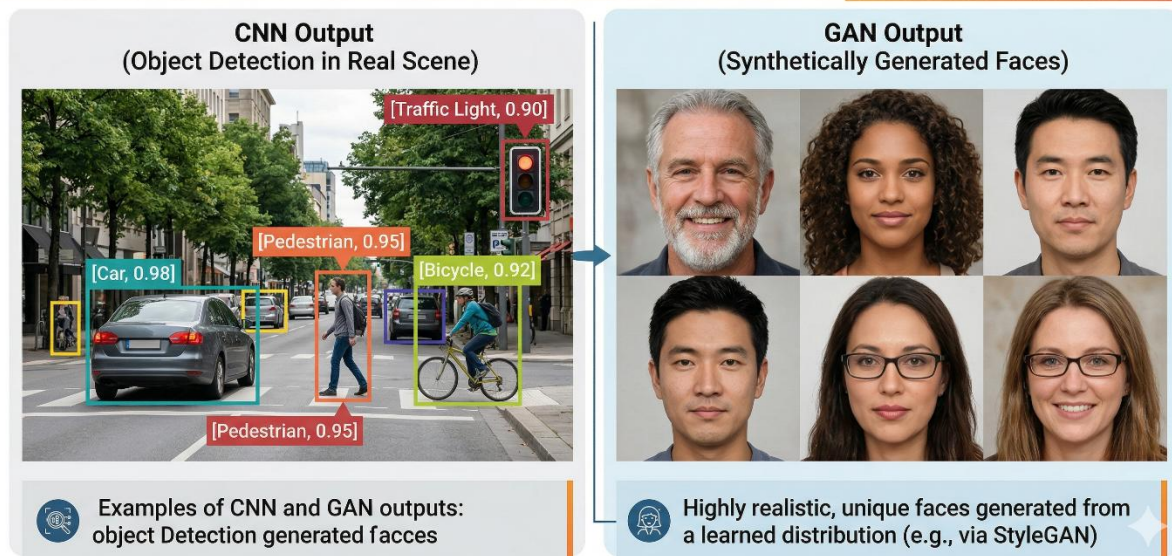
GAN Minimax: $V(D, G) = E[\log D(x)] + E[\log (1 - D(G(z)))]$ (The formula for the competition between Generator G and Discriminator D .)

Strengths and Capabilities

Each architecture is chosen based on its ability to minimize the "Loss Function" for specific tasks.

- **CNN Strengths:** Excellent at image recognition and medical imaging because it recognizes local patterns regardless of position.
- **RNN Strengths:** Superior for Natural Language Processing (NLP) and time-series forecasting because it understands context over time.
- **GAN Strengths:** Unique ability to generate "Actual" looking data from noise, used in deepfakes and high-resolution image upscaling.

EXAMPLES OF CNN OUTPUT (OBJECT DETECTION) AND GAN OUTPUT (GENERATED FACES)



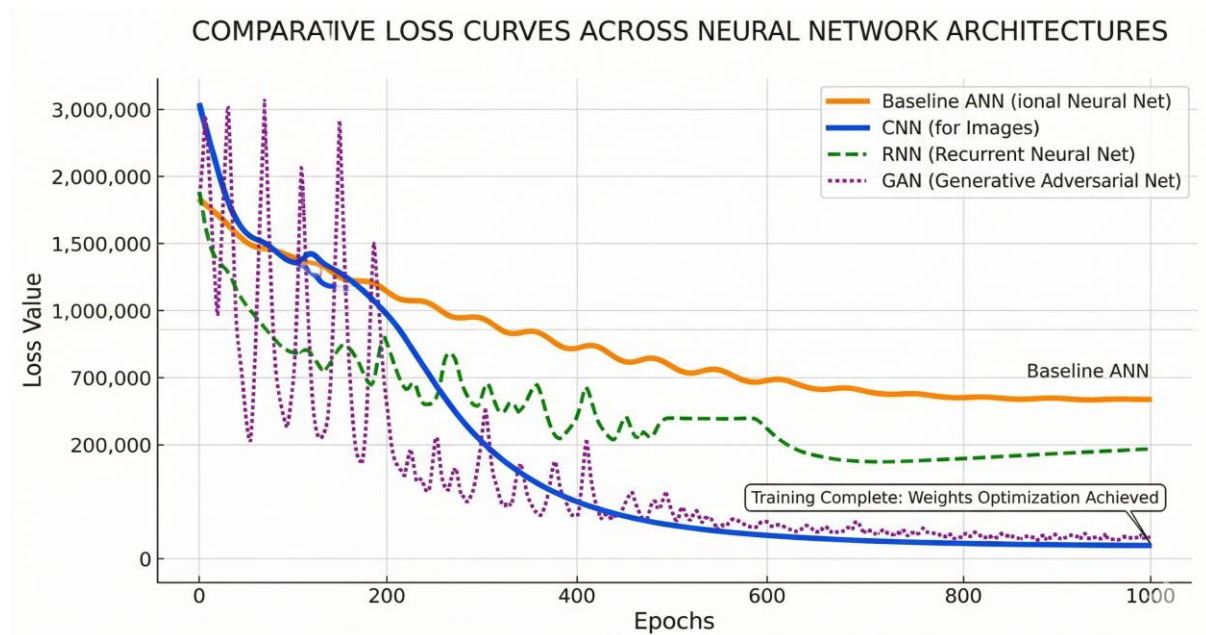
Practical Applications

- **CNN:** Facial recognition, autonomous driving, and X-ray analysis.
- **RNN:** Google Translate, Siri/Alexa voice processing, and stock market prediction.
- **GAN:** Synthetic data generation, clothing design, and photo-to-avatar conversion.

Backpropagation and Learning Errors

Just as you learned in your individual task, these networks use Backpropagation and Gradient Descent to minimize error. However, they face unique hurdles:

- **Vanishing Gradients:** In RNNs, the "Gradient" can become so small that the network stops learning long-term dependencies.
- **Training Instability:** GANs often suffer from "Mode Collapse," where the Generator produces only one type of output, failing to learn variety.
- **Computational Cost:** CNNs require thousands of "Epochs" and high-end GPUs to process large image datasets effectively



Conclusion

By constantly adjusting weights based on error, these architectures simulate different aspects of human intelligence. While CNNs provide the "vision," RNNs provide the "memory," and GANs provide the "imagination" for modern Artificial Intelligence.